

Education

- Vellore Institute of Technology
- Vellore, India

B.Tech Computer Science & Engineering (IoT Specialization) — CGPA: 9.26

2022 - Present

Key Achievements

- National Topper: Responsible & Safe AI Systems (11,876 participants) - IITM & IIITH (2024)
- Top 3 Finalist: Karnataka State Police Hackathon (380 teams) - Demo at Microsoft Research Bangalore (2024)
- Winner: ISRO's IISF-SIF & Samsung PRISM Hackathons (1000+ participants each) (2024)

Technical Skills

- Programming: Python, C++, Java, SQL, JavaScript, LaTeX
- AI/ML: PyTorch, HuggingFace, Scikit-learn, OpenAI Gym, NumPy, Pandas
- GenAI Tools: LangChain, LlamaIndex, vLLM, FAISS, Chroma, OpenAI SDK, Milvus, Qdrant, Google ADK
- Cloud & DevOps: Docker, Kubernetes, Linux, Git, CI/CD, RabbitMQ, Redis, GCP
- Databases & APIs: PostgreSQL, MongoDB, FastAPI, Flask, REST APIs, Microservices
- Key Courses: Algorithms, Data Structures, OOP, Web Programming — View Certifications

Experience

- Samsung MAGPIE R&D Internship

Samsung Research Bangalore

Software Engineering Intern

May 2025 - Jul. 2025

 - Worked with the Server Platform Team and was offered **Pre-Placement Offer(PPO)** based on exceptional performance
 - Designed **asynchronous multi-agentic system** for root cause analysis in distributed systems using Google ADK
 - Built **5 MCP Servers** (Prometheus, K8s, Jaeger, OpenSearch, MilvusDB) for multimodal error analysis & created **dataset of 90+ incidents** using *ChaosMesh* for performance evaluation
- Samsung PRISM R&D Internship

Hybrid

Research Intern

Oct. 2024 - May 2025

 - Architected and implemented **resource-efficient multi-agent system** for **multilingual** Home Automation workflows
 - Developed custom asynchronous orchestration framework with *intelligent task scheduling* and created domain-specific synthetic datasets for cross-lingual agent benchmarking
 - Co-authored research paper *accepted at ICCCNT'25 IIT Indore*
- King Saud University & VIT

Hybrid

ML Research Intern

Dec. 2022 - Present

 - Currently working on **EV placement optimization using reinforcement learning** for large-scale urban infrastructure
 - Developed ML model predicting SoC consumption for electric vehicles with **99.71% accuracy**. *Royal Academy of Engineering* sponsored research
 - Published in **Journal of Energy Storage (IF: 8.9)** - addressing diverse Indian road conditions for EV optimization
- Council of Energy, Environment and Water (CEEW)

CEEW, Delhi

Data Science Intern

Jun. 2024 - Aug. 2024

 - Developed **NLP topic modeling pipeline (BERTopic)** on 92k documents, yielding **350 interpretable topics** for job matching
 - Built ML bias correction algorithms for AQI models, achieving **80% error reduction** - presented at IICAQM'24 IITM
 - Developed solar energy optimization platform with financial modeling (*IRR, payback period, LCOE*) and investment dashboards
- NTNU Norway & BITS Pilani

Remote

AI Research Intern

Mar. 2024 - Aug. 2025

 - Developed **end-to-end OCR system (PyTorch)** using CNN/Transformers (FOTS, Swin), achieving **6% CER** on IAM Dataset
 - Implemented robust processing pipeline with **sliding window architecture** for efficient handwritten text recognition

Key Projects

- Narad: Built **multilingual voice assistant (22 Indian languages)** using **AI4Bharat ASR, FAISS** vector DB for **RAG**, and fine-tuned Llama-2 with contextual bandits
- Sahay: **Gemini Live Voice Agent** for **Mental Health conversational support** using custom ADK Agents (Analysis, Safety) & MCP communication. **Demo — Prototype — Shortlisted R2 Google GenAI Hackathon**
- Aegis: Co-developed **Healthcare** drug delivery system (MIT/Harvard/CERN collab) featuring **federated learning**, multi-agent AI app (MCP), & **RL** for therapy optimization. (*Case study available*)

Publications & Leadership

- Research: Published in **Journal of Energy Storage (IF: 8.9)**; Papers accepted/presented at IICAQM'24, ICCCNT'25, HINT'25, ICDSINC'25
- Leadership: Founder & Team Lead - **Prometheus VIT (RoboCup 2025)**; Chairperson - **Youth Red Cross VIT (780+ blood donations)**